SPLITTING PRINCIPLE: A PROJECTIVE FLAT COVER TRIVIALISES A FINITE COLLECTION OF LOCALLY FREE SHEAVES AND IS INJECTIVE ON CHOW GROUPS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an induction on the total rank, passing to the projective bundle of a sheaf of larger rank and replacing it by the locally free kernel of the canonical surjection

Editorial note. The label lemma-splitting-principle names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The projective bundle formula and the associated canonical surjection onto the tautological quotient with its finite locally free kernel of rank one less
- Injectivity of pullback along the projection of a projective bundle, for every base change
- Filtrations of locally free sheaves with invertible successive quotients, and flat projective morphisms of a given relative dimension

Source reference. Original reference 02TX: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4381–4381. Prerequisite pointer invoked at proof line(s) 5031; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 5005; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $\mathcal{E}_i$ be a finite collection of locally free $\mathcal{O}_X$-modules of rank r_i. There exists a projective flat morphism $\pi : P \to X$ of relative dimension d such that*

(1) *for any morphism $f : Y \to X$ of good algebraic spaces over B the map $\pi_Y^* : \mathrm{CH}_*(Y) \to \mathrm{CH}_{*+d}(Y \times_X P)$ is injective, and*
(2) *each $\pi^*\mathcal{E}_i$ has a filtration whose successive quotients $\mathcal{L}_{i,1}, \ldots, \mathcal{L}_{i,r_i}$ are invertible $\mathcal{O}_P$-modules.*

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Proof. We prove this by induction on the integer $r = \sum r_i$. If $r = 0$ we can take $\pi = \mathrm{id}_X$. If $r_i = 1$ for all i, then we can also take $\pi = \mathrm{id}_X$. Assume that $r_{i_0} > 1$ for some i_0. Let $(\pi : P \to X, \mathcal{O}_P(1))$ be the projective bundle associated to $\mathcal{E}_{i_0}$. The canonical map $\pi^*\mathcal{E}_{i_0} \to \mathcal{O}_P(1)$ is surjective and hence its kernel $\mathcal{E}'_{i_0}$ is finite locally free of rank $r_{i_0} - 1$. Observe that π_Y^* is injective for any morphism $f : Y \to X$ of good algebraic spaces over B, see Lemma 02TX. Thus it suffices to prove the lemma for P and the locally free sheaves $\pi^*\mathcal{E}_i$. However, because we have the subbundle $\mathcal{E}_{i_0} \subset \pi^*\mathcal{E}_{i_0}$ with invertible quotient, it now suffices to prove the lemma for the collection $\{\mathcal{E}_i\}_{i \neq i_0} \cup \{\mathcal{E}'_{i_0}\}$. This decreases r by 1 and we win by induction hypothesis. □